

CURE: Constrained Uncertainty Routing with Explanation-Grounded Experts for Retrieved-Evidence Fact Verification

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Abstract

Fact verifiers often conflate two distinct decisions: which verdict is most likely and whether the retrieved evidence is sufficient to support a decisive verdict. We study this distinction in a fixed-corpus, text-evidence MOCHEG protocol. CURE (Constrained Uncertainty Routing with Explanation-Grounded Experts) combines a verdict-trained direct expert with an explanation-grounded expert. Its *Asymmetric NEI Deferral* rule preserves the direct verdict by default and permits only a high-confidence grounded NEI prediction to defer the output. On frozen validation, this constrained rule improves Macro-F1 from 0.6920 to 0.7112 (+1.92 points; paired-bootstrap $P(\Delta > 0) = 0.9975$), whereas equal averaging and symmetric confidence routing do not. In checksum-audited P1 test inference with frozen $K = 5$ and $\tau = .49$, AND improves Macro-F1 by +0.00977 on raw official test ($n = 2,442$) and +0.00943 on duplicate-safe strict test ($n = 2,434$). The gains concentrate on NEI claims with retrieved evidence available, while retrieval misses remain a failure mode. A separate validation-selected heterogeneous ensemble reaches 0.5551 Macro-F1 on raw P1, and a disagreement-distilled student offers a lower-latency extension.

1 Introduction

Automated fact checking is commonly decomposed into evidence retrieval and claim verification (Guo et al., 2022; Thorne et al., 2018). Modern systems increasingly retrieve heterogeneous evidence and use large language models, but two assumptions remain fragile: that all retrieved passages should be consumed by the verifier, and that a model trained to predict a verdict is also a reliable detector of *retrieved-evidence sufficiency*. We focus on the latter distinction in a controlled text-evidence setting.

MOCHEG (Yao et al., 2023) provides a useful setting because it couples three-way truthfulness

labels with textual, visual, and social evidence. We deliberately isolate its text-evidence component: all experts receive the same fixed-corpus retrieved passages, so that effects of supervision and decision policy are not confounded by live-web retrieval or visual modeling. This is therefore a fact-verification study on a multimodal benchmark, rather than a claim that the proposed system itself is multimodal.

Our central observation is that two verifiers have complementary error profiles. A direct verifier is strong when evidence appears decisive but can overcommit under non-decisive retrieved context. An explanation-grounded expert is more likely to predict NEI, but is not uniformly more accurate. We turn this complementarity into a constrained policy rather than averaging the experts. The direct expert remains the default; only a high-confidence grounded NEI signal can defer the decision. We call this operator *Asymmetric NEI Deferral* (AND). It is a class-constrained cross-expert decision rule, not Bayesian epistemic uncertainty estimation.

Our contributions are:

- CURE, a dual-expert formulation that separates direct verdict prediction from explanation-grounded retrieved-evidence sufficiency signals;
- AND, a class-constrained cross-expert rule that permits only NEI deferral, and controlled ablations against grounded-only, posterior averaging, and symmetric routing alternatives;
- an analysis showing that the benefit is concentrated in NEI cases with high-quality retrieved evidence, not a universal improvement across labels or retrieval conditions; and
- a disagreement-distilled student as a secondary, lower-latency deployment extension.

We distinguish confirmatory results from development diagnostics. The heterogeneous ensemble and AND answer different questions: the former tests broad probability complementarity, while the latter tests a registered, class-constrained decision policy. Both are evaluated on their respective frozen test protocols without test-time parameter selection.

2 Related Work

Fact verification and evidence sufficiency. FEVER established evidence-supported three-way verification at scale (Thorne et al., 2018); LIAR studied short political claims (Wang, 2017); and MultiFC broadened evaluation across sources and domains (Augenstein et al., 2019). These benchmarks make clear that label prediction and evidence sufficiency are not identical. Surveys consequently treat retrieval, evidence selection, reasoning, and calibration as coupled but distinct stages (Guo et al., 2022). UKP-Athene illustrates the role of multi-sentence textual entailment in evidence-based verification (Hanselowski et al., 2018).

Retrieval, selection, and reasoning. Evidence retrieval can dominate downstream verification (Zheng et al., 2024). RAV combines retrieval with verification, while RAFTS synthesizes contrastive arguments from retrieved evidence (Yue et al., 2024). Complex-claim verification decomposes claims before retrieving web evidence (Chen et al., 2024a). These approaches study different evidence regimes from our P1 protocol: they motivate the separation between fixed-corpus evidence selection and live-web retrieval rather than serving as directly interchangeable leaderboard rows.

MOCHEG and multimodal fact checking. MOCHEG introduced end-to-end multimodal fact checking and explanation generation (Yao et al., 2023). Recent systems pursue different ways to aggregate textual and visual evidence: LVL4M4FV uses large vision-language models with reranked evidence (Tahmasebi et al., 2024); MetaSumPerceiver creates claim-specific multimodal summaries (Chen et al., 2024b); HGTMF models high-order interactions with hypergraphs (Pang et al., 2025); MEVER jointly addresses retrieval, verification, and explanations (Zhang et al., 2026); and M-RAV augments cross-modal evidence alignment for zero-shot verification (Luu et al., 2027). AMuFC adaptively determines whether visual evidence is needed (Jung et al., 2026). These sys-

tems motivate the benchmark comparison, but our P1 method consumes retrieved text only. CURE consequently studies constrained routing between differently trained text verifiers, not multimodal fusion.

3 Task Definition and Protocols

Given a claim c and retrieved evidence $E = \{e_j\}_{j=1}^K$, predict $y \in \{\text{SUPPORTED}, \text{REFUTED}, \text{NEI}\}$. The principal P1 protocol uses MOCHEG’s fixed corpus and upstream text retrieval; it is not an open-web evaluation. The raw official test contains 2,442 claims. A strict robustness track removes eight verbatim cross-split duplicates and contains 2,434 claims. We report both tracks separately. Development validation contains 1,456 claims.

All routing thresholds are selected on validation or train-only out-of-fold data. Test labels are never used for selecting a seed, threshold, or policy in the confirmatory result. Open-web experiments in Phase C are not mixed into P1 tables.

4 Method

4.1 Direct and grounded experts

The direct expert f_D receives a claim c and the top- K reranked passages E_K and predicts a three-way distribution $p_D(y \mid c, E_K)$. It is optimized only with verdict cross-entropy. The grounded expert f_G has the same Qwen3-4B-LoRA backbone and the same inference prompt, but is additionally trained to generate a structured teacher rationale. Thus, it remains a verdict classifier at inference rather than a rationale generator.

Teacher rationales are generated once for training claims by Qwen2.5-7B-Instruct using the *gold training verdict* and the system-retrieved top-five passages. The teacher is instructed to mark a rationale as ungrounded when those passages cannot justify that training verdict. A valid target contains a grounded flag, verdict, key-passage identifiers, free-text reason, and missing-information field. Generation is deterministic (greedy decoding, maximum 512 new tokens), and schema validation rejects malformed key-passage references and unsupported structured outputs. Explanation loss is applied only to valid, grounded targets. This use of training labels is ordinary supervised learning; no validation or test label, and no gold evidence passage, is included in a validation/test prompt.

For a training example i , let z_i be the verdict target and r_i the structured rationale target. The two expert objectives are

$$\begin{aligned}\mathcal{L}_D &= - \sum_i \log p_D(z_i | c_i, E_i), \\ \mathcal{L}_G &= \mathcal{L}_D + \lambda_r \mathcal{L}_{\text{rationale}}(r_i), \quad \lambda_r = 0.25.\end{aligned}\tag{1}$$

The rationale term is token-level cross-entropy over the validated structured target. Both experts begin from the same base model, but the headline direct and grounded ensembles were trained under different optimization recipes (specified below); therefore their contrast establishes complementarity, not a causal attribution of every gain to rationale loss. We additionally report a matched-recipe explanation-supervision control and treat it as a causal ablation rather than conflating it with the headline ensemble comparison. The implementation records teacher outputs, manifests, retrieval signatures, and prompt-format hashes. Gold evidence is used only for retrieval diagnostics, never injected into P1 validation or test prompts.

4.2 Asymmetric NEI deferral (AND)

The policy uses the direct expert as an anchor:

$$\hat{y} = \begin{cases} \text{NEI}, & \arg \max_y p_G(y) = \text{NEI} \wedge p_G(\text{NEI}) \geq \tau, \\ \arg \max_y p_D(y), & \text{otherwise.} \end{cases}\tag{2}$$

We use $\tau = 0.49$, selected on validation and frozen thereafter for the registered $K = 5$ router. This rule is intentionally asymmetric: the grounded expert is not permitted to overwrite a direct Supported or Refuted decision with another decisive class. It can only defer to NEI when its retrieved-evidence insufficiency signal is high. AND is therefore distinct from posterior averaging, symmetric confidence routing, and an unconstrained expert switch. We use “sufficiency” operationally: $p_G(\text{NEI})$ is a learned class posterior, not a Bayesian estimate of parameter uncertainty.

4.3 Disagreement-aware distillation

B19 is a secondary single-model variant. Let q_D and q_G be frozen teacher distributions generated without the held-out fold. For agreement cases, the distillation target is their average; for disagreement cases, a train-only cross-fitted outcome identifies which teacher target to transfer. The confirmed B19 variant minimizes

$$\mathcal{L}_{\text{DKD}} = \mathcal{L}_{\text{CE}} + \lambda_{\text{KD}} T^2 \text{KL}(q \| p_S).\tag{3}$$

Split/protocol	Claims	Labels used for selection	Role
Train	11,631	train only	optimization
Validation	1,456	validation	frozen policy
Official test	2,442	no	selection
Strict test	2,434	no	primary benchmark
			duplicate-robustness

Table 1: Protocol sizes. Official and strict test results are not pooled.

Here q is the selected teacher target and $T = 2$ is the distillation temperature. A counterfactual sufficiency loss was tested during the Fold-0 mechanism screen but was not retained in the confirmed disagreement-KD variant. B19 is reported as a compute-efficient secondary extension, not as a replacement for AND.

4.4 Competitive ensemble variant

The CURE-Ensemble test system is a complementary, validation-selected ensemble: five frozen direct seeds plus the three best grounded seeds ranked on validation. The composition $\{100, 87, 42\}$ is fixed before the official test is evaluated, and its prediction is the arithmetic mean of member verdict probabilities. We also report an unpruned ten-member ensemble (all five direct plus all five grounded seeds) as a no-subset-selection control. This system is a competitive benchmark variant, not an implementation of AND: it answers whether heterogeneous expert predictions are complementary, whereas AND answers whether a constrained NEI deferral is useful.

5 Experimental Setup

5.1 Data and preprocessing

MOCHEG contains 15,601 English claims from PolitiFact and Snopes with textual, visual, and social evidence (Yao et al., 2023). Our P1 experiments use a fixed MOCHEG *text* corpus. Qwen3-Embedding-4B retrieves a top-50 dense candidate pool, and Qwen3-Reranker-4B then scores claim–passage utility with an instruction requiring entity, temporal, quantity, location, and negation compatibility. The registered router consumes the top five reranked passages ($K = 5$), truncated to 2,200 evidence characters. No gold evidence is injected at validation or test. The open-web C-series snapshot is intentionally excluded from P1 effectiveness tables because it is a different deployment regime.

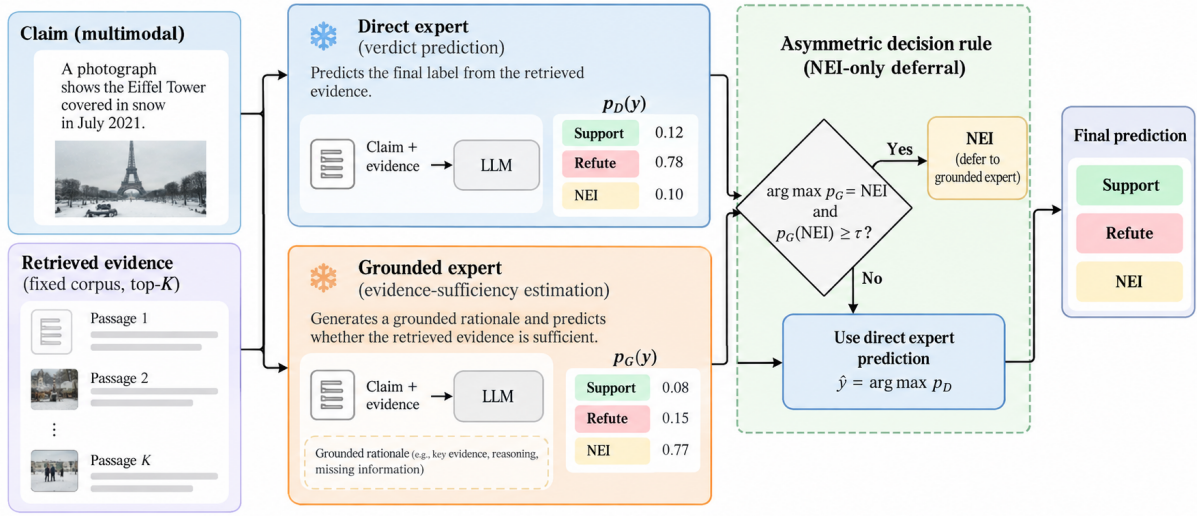


Figure 1: Overview of CURE. The claim and retrieved evidence are processed by (i) a direct verdict expert and (ii) an explanation-grounded expert. AND may only defer a decision to NEI; otherwise it preserves the direct prediction. The lower B19 branch is a secondary distillation extension, not a required component of AND.

5.2 Models and metrics

All verifiers use Qwen3-4B-Instruct-2507 with LoRA adapters. The direct ensemble uses seeds $\{13, 21, 42, 87, 100\}$; the grounded ensemble contains the validation-selected seeds $\{100, 87, 42\}$. The representative B1 runs use LoRA rank 16, LoRA scaling 32, dropout 0.05, learning rate 10^{-4} , AdamW weight decay 0.01, warm-up ratio 0.05, three epochs, per-device batch size 1, and gradient accumulation 16. The headline explanation-grounded runs use the same LoRA rank/scaling/dropout but batch size 2, accumulation 4, and learning rate 2×10^{-4} . Hence their raw ensemble comparison is not a matched-optimization causal ablation. For causal attribution, we separately use the five-fold matched-recipe control reported in Section 7.3. Exact run configurations are retained in JSON summaries.

We report accuracy, class F1, Macro-F1, expected calibration error (ECE) when available, paired percentile bootstrap intervals, and exact McNemar tests. Bootstrap samples are drawn at the claim level with 10,000 iterations. We call $P(\Delta > 0)$ a bootstrap probability of superiority, not a frequentist p -value. All test table deltas compare predictions with matched claim IDs.

5.3 Baselines and comparison discipline

Baselines include direct-only, grounded-only, equal posterior averaging, symmetric confidence rout-

ing, no-asymmetric-NEI routing, disagreement-only routing, max-confidence routing, and oracle routing. Literature scores are reported in a separate protocol-aware table. MOCHEG, HGTMFC, AMuFC, MEVER, M-RAV, and open-web systems do not necessarily share evidence, split, modality, or metric definitions; therefore a row is not treated as a direct leaderboard comparison unless its protocol is explicitly aligned.

5.4 Promotion gates and split discipline

Every development intervention is governed by a predefined promotion gate stored with the corresponding run configuration. From B6 onward, the project uses duplicate-family-safe train-only folds, fixed training epoch, compute-matched controls, and fresh confirmation folds. A screen-only gain therefore cannot become an official-test claim. CURE-Ensemble is selected on validation and applied once to test. The post-hoc $\tau = .60$ sensitivity result is reported separately and never used as a confirmatory result. The registered AND condition uses $K = 5$ and $\tau = .49$; a later K sweep is descriptive rather than a retrospective change to this condition.

6 Main Results

6.1 Mechanism-focused evidence

We report only the development evidence needed to assess the proposed mechanism in the main paper.

Development question	Confirmed observation
Evidence compression	Atomic evidence loses discourse context; a rationale-distilled selector does not exceed retrieval Top-3.
Symmetric multi-task learning	B6 auxiliary transfer and PCGrad fail their fresh-fold or bootstrap promotion gates.
Calibration and routing	Logit bias, cross-fit calibration, and unconstrained routers produce small, non-robust gains.
Counterfactual supervision	A +.01654 Fold-0 screen gain becomes -.00104 versus matched control in confirmation.

Table 2: Four negative-result families motivating the constrained design. Detailed phase-level results are in Appendix A.

Variant	Acc.	Macro-F1	F1-S	F1-R	F1-NEI
A0 Direct-only	.700549	.692045	.720348	.769904	.585882
A1 Grounded-only	.706731	.701191	.704519	.780834	.618221
A2 Equal posterior average	.699863	.692773	.707071	.772608	.598639
A3 Symmetric confidence switch	.698489	.691095	.706145	.771930	.595211
A4 Unconstrained confident switch	.710165	.704436	.707904	.781915	.623491
CURE (AND)	.714973	.711205	.703016	.793008	.637591

Table 3: Frozen validation component ablations (validation is used only to evaluate fixed policies). “Unconstrained confident switch” selects the grounded output of any class when its maximum probability is at least τ . CURE applies the class-constrained AND rule at $\tau = .49$. Degenerate disagreement-only and max-confidence variants are omitted from the main table and reported in the released ablation artifact. F1-S/R denote Supported/Refuted.

Earlier B2–B17 experiments are retained as a compact appendix ledger. They establish four negative findings: evidence compression did not reliably replace document context; symmetric auxiliary objectives and PCGrad did not generalize; calibration and unconstrained routing alternatives were unstable; and counterfactual curricula failed matched-control confirmation. These failures motivate retaining a competent direct verifier and treating the grounded branch as a constrained NEI signal rather than a universally superior classifier.

6.2 Frozen validation and ablations

Broader prior-work inventory. Table 4 lists twelve relevant MOCHEG system rows. It is deliberately *not* a single leaderboard: a reported F-score is preserved as F-score rather than silently renamed Macro-F1; filtered tests, gold evidence, or alternate preprocessing are marked. Only rows labelled “aligned raw P1” support an apples-to-apples point-estimate comparison.

AND is the strongest frozen validation policy. Relative to direct-only it gains 1.916 Macro-F1 points and 1.442 accuracy points. Replacing the

class-constrained rule by an unconstrained confident switch reduces Macro-F1 by 0.677 points, whereas equal averaging loses 1.843 points. The grounded-only branch is useful (+0.915 over A0) but remains 1.001 points below AND. The independently inferred raw and strict P1 results below retain the direction of this effect, while their uncertainty intervals require a cautious interpretation.

6.3 Official test results and leakage control

The frozen CURE-Ensemble reaches Macro-F1 0.55507 and accuracy 0.57535 on raw P1 official test, compared with 0.54531 and 0.56798 for the direct five-seed ensemble. The paired bootstrap probability of improvement is 0.9839 with CI [+0.00097, +0.01850]. Its score is higher than AMuFC’s reported raw-P1 point estimate (Macro-F1 .5400; accuracy .5460), but this is not a paired comparison and the two systems differ in retrieval, backbone, and modality use. We therefore report it as a protocol-aligned point estimate rather than a claim of statistical superiority or state of the art. CURE-Student disagreement distillation reaches Macro-F1 0.54940 with one student seed; it is an efficiency diagnostic, not a multi-seed official-test comparison.

AND now has canonical test artifacts on both tracks. On raw P1, AND obtains .55470 Macro-F1 and .57125 accuracy, compared with its independently inferred direct anchor’s .54493 and .56798: $\Delta\text{MF1} = +.00977$, with 86 helpful and 78 harmful changes and paired bootstrap $P(\Delta > 0) = .9546$. On strict P1, it obtains .55433 versus .54490 ($\Delta\text{MF1} = +.00943$; 84/77 helpful/harmful; $P(\Delta > 0) = .9485$). The raw result passes the preregistered probability-of-superiority threshold by a narrow margin; the strict result is consistent in point estimate but narrowly below that probability threshold. The percentile intervals include zero on both tracks, and McNemar tests are non-significant, so we treat this as confirmatory but modest evidence rather than a claim of large or unqualified statistical superiority. Both reports record manifest, retrieval, input-prediction, and materialized-output hashes. Because raw and strict ranked evidence differ for some shared IDs, each track is inferred independently from the same frozen checkpoints; no intersection or test-time tuning is used.

System	Venue/year	Evidence and evaluation	Acc.	Reported F1	Comparability
MOCHEG (Yao et al., 2023)	SIGIR 2023	system text+image	–	F-score .4406	original benchmark; score label retained
LVL4M4FV (Tahmasebi et al., 2024)	arXiv 2024	retrieved text+image	.4510	micro-F1 .4510	macro not source-reported
HGTMFC (Pang et al., 2025)	AAAI 2025	retrieved text+image, raw test	.4861	Macro-F1 .4678	aligned split; different model/retrieval
MetaSumPerceiver (Chen et al., 2024b)	ACL 2024	system text+image	–	F-score .4860	source does not label it
MEVER (Zhang et al., 2026)	arXiv 2026	retrieved multimodal	.483±.021	MF1 .497±.012	Macro-F1 custom consistent preprocessing
M-RAV (Qwen2.5-32B) (Luu et al., 2027)	IPM 2027	system evidence, filtered $n = 2001$	.5002	MF1 .5014	filtered test; not aligned
Entailed Opinion (Kumar et al., 2025)	arXiv 2025	domain-generalization setting	–	MF1 .5700	not P1 system retrieval
AMuFC (Jung et al., 2026)	HCAI/CIKM 2025; arXiv 2026	retrieved text+image, raw test	.5460	MF1 .5400	aligned raw P1 challenger
CURE Direct ensemble	this work	fixed corpus, raw P1 $n = 2442$	.5680	MF1 .5453	aligned raw P1
CURE Grounded seed 100	this work	fixed corpus, raw P1 $n = 2442$	.5696	MF1 .5513	aligned raw P1, one seed
CURE AND (5+3 experts)	this work	fixed corpus, frozen $K = 5, \tau = .49$, raw P1	.5713	MF1 .5547	aligned raw P1, canonical router
CURE-Ensemble (8-model)	this work	validation-selected, raw P1 $n = 2442$	.5754	MF1 .5551	aligned raw P1, confirmatory variant

Table 4: Protocol-aware comparison with prior work. “Aligned” means raw MOCHEG P1 test with system-retrieved evidence; it does not assert identical retrievers or model capacity. The table has more than ten prior/system rows, but only aligned rows may be used for a direct point-estimate comparison.

Evidence policy	Acc.	Grounded MF1	AND MF1
$K = 1$	.707418	.660969	.704452
$K = 3$	.718407	.695423	.715270
$K = 5$	.714973	.701191	.711205
Direct-only	.700549	–	.692045

Table 5: Evidence-count sensitivity on validation. The registered AND condition uses $K = 5$, inherited from the B1 retrieval protocol. The later $K = 3$ sweep has the highest validation score but is not substituted for the registered condition or used for an official-test claim.

τ	Validation MF1	Status
.40	.70774	sensitivity
.45	.70932	sensitivity
.49 (frozen)	.71121	registered AND
.50	.71109	sensitivity
.55	.71089	sensitivity
.60	.71012	sensitivity only
.65	.70755	sensitivity
.70	.70420	sensitivity

Table 6: Validation threshold sensitivity for AND. $\tau = .49$ is frozen for the registered condition. No raw-test threshold sweep is used for parameter selection.

CURE system	Macro-F1	Accuracy	Raw-P1 status
B1 five-seed	.5453	.5680	frozen baseline
CURE Grounded seed 42	.5338	.5524	one seed
CURE Grounded seed 87	.5394	.5606	one seed
CURE Grounded seed 100	.5513	.5696	best single grounded seed
CURE-Ensemble full 10-member	.5512	.5717	unpruned control
CURE-Ensemble selected 8-member	.5551	.5754	confirmatory benchmark variant
CURE AND (5+3 experts)	.5547	.5713	canonical frozen router
CURE-Student DKD seed 87	.5494	.5676	raw P1, one seed

Table 7: Raw official P1 test ($n = 2,442$) for frozen CURE system variants. CURE-Ensemble and AND use distinct, validation-frozen compositions/policies and should not be read as interchangeable ablations. AND uses registered $K = 5, \tau = .49$ with independently inferred raw-P1 predictions.

Evidence policy	Acc.	MF1	F1-S	F1-NEI
Original Top-5 anchor	.6664	.6435	.6026	.5067
Retrieval Top-3	.6720	.6482	.5964	.5187
Generic CE selector	.6496	.6374	.5674	.5253
Rationale-distilled selector	.6651	.6460	.6098	.5103

Table 8: B18B-Sel train-only Fold-0 screen ($n = 2,326$). The rationale-distilled selector is better than generic CE but does not exceed retrieval Top-3.

7 Additional Ablations and Negative Results

7.1 Sentence-selection branch

A separately named internal experiment, *B18B-Sel*, distills teacher key-passage attribution into a MiniLM cross-encoder that adaptively selects one to three sentences. It is an evidence-selection ablation, not the dual-expert router. On Fold 0, rationale-distilled selection improves a generic cross-encoder by +.00865 MF1 but remains -.00214 behind the retrieval Top-3 control. It was

therefore closed after its predefined screen rather than expanded to more seeds or test evaluation (Table 8). This result prevents us from attributing the grounded-expert effect to sentence compression alone; it does not isolate the causal contribution of teacher capacity versus structured-rationale supervision.

7.2 Why earlier constraint approaches were not promoted

Earlier B6–B17 interventions supplied useful diagnostics but did not establish a stable main method. B6-A’s raw auxiliary ensemble reached .7080 MF1 on validation, yet its bootstrap probability against the compute-matched direct control was .9354, below the predefined .95 gate. B9 and B13 exhibited strong Fold-0 gains but degraded under fresh-fold confirmation. B16’s counterfactual verdict curriculum improved Fold 0 by +.01654 MF1 versus its anchor, but aggregated confirmation was -.00104 versus the matched control. These negative outcomes motivate the CURE design choice: preserve a competent direct decision model and use grounded supervision as a constrained retrieved-evidence signal rather than continuously modifying the anchor’s decision boundary.

7.3 Matched-recipe explanation supervision control

To separate the rationale objective from training exposure, we also ran a five-fold control in which verdict-only and explanation-supervised models use the same B18 training script and optimizer schedule. The explanation model obtains .65757 aggregate MF1 versus .65762 for the matched verdict-only control ($\Delta = -.00006$; bootstrap $P(\Delta > 0) = .4940$). Applying the frozen asymmetric router to this matched explanation expert yields .65898, only +.00135 over the control ($P(\Delta > 0) = .7273$). Thus, the available matched control does *not* establish that structured rationale loss alone improves average verdict accuracy. We retain the explanation-grounded branch because it yields a complementary NEI-oriented signal under the registered AND interface, but make no stronger causal claim about the source of that complementarity.

8 Efficiency and Robustness Analysis

8.1 Inference efficiency

On an RTX 5090 with 256 common validation claims, batch size 4, and two warm-up batches, direct, grounded, AND, and the one-model student require 320.72, 193.17, 513.88, and 64.47 ms per claim, respectively (Table 9). The student uses one pass instead of the router’s eight sequential passes. All ensemble timings are sequential; the repeated 14.74 GiB figure is the maximum allocated memory of one loaded member, not a comparison

System	Passes	ms/claim	claims/s	Peak GiB
Direct ensemble	5	320.72	3.118	14.74
Grounded ensemble	3	193.17	5.177	14.74
AND dual ensemble	8	513.88	1.946	14.74
CURE-Student	1	64.47	15.511	14.74

Table 9: Efficiency benchmark: RTX 5090, 256 claims, batch size 4, two warm-up batches. Model-loading time is excluded.

Group	n	Δ MF1	Helpful/Harmful
Overall	1,456	+0.01916	58 / 37
PolitiFact	431	-.00249	0 / 6
Snopes	1,025	+0.01994	58 / 31
Gold NEI	478	+0.03239	58 / 0
Gold Refuted	487	-.00343	0 / 9
Gold Supported	491	-.01404	0 / 28
Gold rank 1	1,233	+0.02772	54 / 25
Gold rank 2–5	131	-.02607	3 / 7
Gold not retrieved	59	-.03260	1 / 4

Table 10: Selected frozen-validation subgroup audit: AND minus direct. Gold rank is a diagnostic label unavailable at inference, not a routing input.

of total ensemble memory. Subset effectiveness scores produced by this benchmark are diagnostic only.

8.2 Subgroup behavior

On the frozen validation audit, AND improves Macro-F1 by 1.916 points, with 58 helpful and 37 harmful changes. Table 10 separates where this gain occurs. It is largest for NEI (+3.239), rank-one evidence (+2.772), evidence hits (+2.280), and Snopes (+1.994). Retrieval misses decrease by 3.223 points and Supported claims decrease by 1.404 points. These results support only a narrow interpretation: the grounded branch is useful when retrieved evidence is available and non-decisive in a familiar pattern. They do not establish that it detects corpus-level insufficiency under retrieval failure.

8.3 B19 distillation

Five-fold train-only OOF confirmation gives the B19 control Macro-F1 0.660788 and disagreement-KD Macro-F1 0.667607, a +0.006819 difference. Four of five folds improve and paired bootstrap $P(\Delta > 0) = 0.9517$. Table 11 shows why the disagreement target matters: naive ensemble KD is worse than the matched control, while counterfactual loss alone is nearly neutral and the full objective reduces the screen gain. This supports B19 as a practical compression extension, but its margin is close to the predefined bootstrap threshold and its official-test evidence is only a single seed.

Fold-0 variant	Acc.	MF1	F1-S	F1-NEI
Matched control	.6690	.6570	.6013	.5368
Naive ensemble KD	.6707	.6513	.6106	.5152
Counterfactual only	.6758	.6577	.6063	.5332
Full KD + disagree + CF	.6776	.6584	.6022	.5359
Disagreement KD	.6836	.6687	.6276	.5472

Table 11: B19 train-only Fold-0 mechanism screen ($n = 2,326$). The five-fold confirmation uses disagreement KD only; counterfactual terms are reported solely as screen ablations.

9 Error Analysis

The dominant helpful transition is direct Refuted/Supported to grounded NEI on gold NEI claims. Harmful transitions are direct Supported to NEI (28 cases) and direct Refuted to NEI (9 cases). This explains why AND improves NEI while reducing decisive-class F1. The failure atlas also shows that routing under missing or rank-2–5 evidence is less reliable. A future system should estimate retrieval reliability before allowing deferral, but using gold rank or QREL status at inference would be invalid.

10 Discussion

The results support a narrow interpretation of the contribution. Explanation-grounded training creates a complementary retrieved-evidence NEI signal, but explanation supervision alone is not consistently beneficial: a matched explanation-loss ablation produced approximately zero aggregate gain. The useful object is the asymmetric interface between a sufficiency-oriented signal and verdict prediction. This interface also exposes a compute-quality trade-off: AND uses eight sequential expert passes, whereas B19 compresses complementary teacher behavior into one student. A direct-expert NEI-threshold baseline and a dedicated sufficiency-detection AUROC/AUPRC study remain necessary before claiming that the grounded probability is a calibrated insufficiency detector.

11 Limitations and Ethical Considerations

MOCHEG is a single benchmark with two principal sources and potentially duplicated claims; we therefore report raw and strict tracks. The fixed-corpus protocol does not measure live-web robustness or establish multimodal effectiveness. The grounded expert inherits teacher errors and pseudo-label noise; moreover, teacher rationales are condi-

tioned on the gold *training* verdict, which is appropriate for training but prevents the rationale channel from being interpreted as independent supervision. Subgroup analyses are descriptive and do not justify post-hoc policy tuning. Although AND now has independently inferred, checksum-audited raw and strict test artifacts, its 95% paired-bootstrap intervals include zero and its strict-track superiority probability is .9485; stronger multi-dataset confirmation is necessary before broad generalization claims. A direct NEI-threshold baseline, a teacher-capacity ablation, and an external dataset remain necessary before submission. Automated fact checking can amplify source and retrieval bias; predictions should be treated as decision support, not authoritative adjudication.

12 Conclusion

We presented CURE, a dual-expert constrained routing framework for retrieved-evidence fact verification. Its AND rule improves frozen-validation Macro-F1 by 1.92 points and improves independently inferred raw/strict P1 test Macro-F1 by .977/.943 points under a frozen $K = 5, \tau = .49$ policy. The effect is modest and uncertainty-aware: only raw P1 clears the preregistered .95 bootstrap-probability gate. CURE-Ensemble provides a separate validation-selected benchmark variant, while B19 demonstrates a one-model compression extension. The next research steps are external-dataset confirmation and sharper retrieval-reliability gating, rather than post-hoc policy tuning on P1.

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A Detailed Development Ledger

Table 12 expands the main trajectory with the numerical evidence used to close or advance each hypothesis. These are reported for reproducibility and to prevent failed development screens from being misrepresented as official-test improvements.

B Five-Fold B19 Confirmation

C Raw versus Strict Test Reporting

The raw official test ($n = 2,442$) is the primary literature-comparison track. The strict test ($n = 2,434$) removes eight verbatim cross-split duplicates and is a robustness track only. Table 14 does not merge the two populations. Canonical AND uses independently inferred predictions on each retrieval manifest because their ranked evidence IDs differ for some shared claims; every prediction file is checked against its full manifest rather than evaluated on an intersection.

D Reproducibility Checklist

The repository records manifests, retrieval signatures, prediction files, training summaries, seeds, and protocol flags. The main scripts are ‘train_mocheg_qwen3_lora_verifier.py’, ‘train_mocheg_b18_explanation_verifier.py’, ‘analyze_mocheg_b18b_component_ablations.py’, ‘analyze_mocheg_b18b_subgroups.py’, ‘train_mocheg_b19_distillation.py’, ‘benchmark_mocheg_b18b_efficiency.py’, and ‘evaluate_mocheg_b18b_canonical_router.py’. Before submission, add exact GPU driver versions, package-lock information, author metadata, and a public archival repository link.

ID	Intervention	Quantitative observation	Interpretation
B2-SV	flat versus hierarchical sufficiency verification	Flat MF1=.6371; SV=.6398 (+.0027) on Fold 0; five-fold confirmation $+.00060 \pm .00745$	Hierarchy had small local benefit but no stable promotion.
B3	source/evidence DRO	qrel-absent MF1 +.1004; overall MF1 -.0021	Robustness improvement cannot be bought with aggregate regression.
B4	atomic sentence verifier	sentence retrieval Recall@8=.8661; verdict MF1 -.0270 vs article evidence	Retrieval coverage alone is insufficient; discourse matters.
B5	context packets	best interpolation $+.01014$ on one validation split; five-seed packet gain $+.00387 \pm .00862$	Complementarity exists but is not a reliable replacement.
B6-A	sufficiency/polarity multi-task	auxiliary vs matched control $+.00886 \pm .01081$; bootstrap .9354	Positive transfer signal, below preregistered stability gate.
B6-B	PCGrad	conflict rate about 45–46% in epoch 1; ensemble -.0105 vs control	Gradient conflict was real but projection did not improve generalization.
B7	selective expert router	MF1 $+.00460$; bootstrap .9294	Helpful precision, insufficient gain and confidence.
B8	class-logit bias	MF1 $+.00216$; bootstrap .7604	Prior shift does not explain the main error pattern.
B9	fixed seed ensemble	Fold-0 $+.01348$; fresh-fold mean $+.00398$	A screen peak is not confirmation.
B10	cross-fitted calibrator	aggregate MF1=.67054; $+.00345$ vs unweighted ensemble	Bootstrap/source gates fail.
B11	provenance calibrator	-.00176 vs unweighted ensemble	Provenance-conditioned calibration harms aggregate accuracy.
B12	joint constraints from base	$+.00538$ vs compute control but $-.01376$ vs anchor	Constraint tasks help relative to compute but induce anchor drift.
B13	curriculum/blend	Fold-0 blend $+.00510$; five-fold mean $+.00090$	Direct curriculum is unstable.
B14	NEI escape analysis	MF1 $+.00209$; accuracy $+.01126$	It helps supported/refuted escape but damages NEI under qrel absence.
B15	value-of-information gate	MF1 $+.00380$; AUROC=.5660	Observable utility is too weak for reliable routing.
B16	counterfactual omission	Fold-0 $+.01654$ vs anchor; confirmation $-.00104$ vs control	The strongest screen still fails matched-control confirmation.
B17	failure atlas	candidate/control .66517/.66621; help/harm 550/561	No stable subgroup justified further tuning on inspected folds.

Table 12: B2–B17 negative and diagnostic experiments. Each value remains in its native protocol; no row is a P1 official-test claim.

Fold	Control MF1	DKD MF1	Δ
0	.657035	.668656	$+.011621$
1	.659998	.656767	$-.003231$
2	.662265	.667143	$+.004878$
3	.647130	.660188	$+.013058$
4	.674931	.681645	$+.006714$
Aggregate	.660788	.667607	$+.006819$

Table 13: B19 train-only out-of-fold confirmation. Four of five folds improve; the aggregate paired bootstrap probability is .9517.

System	Track	Acc.	MF1
B1 five-seed	raw official	.56798	.54531
B1 five-seed	strict	.56902	.54581
B18A CURE-Ensemble	raw official	.57535	.55507
B18A CURE-Ensemble	strict	.57477	.55383
B19 seed 87	raw official	.56757	.54940
B19 seed 87	strict	.56615	.54733
B18B AND router	raw official canonical	.57125	.55470
B18B AND router	strict canonical	.57149	.55433

Table 14: Track-separated test results. B18B AND uses frozen $K = 5, \tau = .49$ and independently inferred raw/strict predictions; it is not an intersection evaluation.